Journal of
Materials Chemistry A

PAPER

Cite this: *J. Mater. Chem. A*, 2018, **6**, 11951

Aluminum-incorporated p-CuO/n-ZnO photocathode coated with nanocrystal-engineered TiO_2 protective layer for photoelectrochemical water splitting and hydrogen generation†

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Received 2nd April 2018
Accepted 31st May 2018

DOI: 10.1039/c8ta03027h

rsc.li/materials-a

The poor photocorrosion stability and low photovoltage of cupric oxide (CuO) are the main limiting factors of CuO-based photocathodes for solar-driven photoelectrochemical (PEC) water splitting and hydrogen evolution. In this paper, we demonstrate a highly efficient CuO-based photocathode fabricated on a glass substrate coated with fluorine-doped tin oxide (FTO). Incorporating aluminum (Al) into the thin CuO film (CuO:Al) and inserting a thin CuO interfacial layer between the CuO:Al film and the FTO-coated glass substrate is shown to improve the photocorrosion stability of the CuO and increase the photocurrent without increasing the dark current. We also demonstrate that depositing a layer of ZnO to form a buried p-(CuO/CuO:Al)/n-ZnO:Al heterojunction and controlling the carrier concentration and conductivity of the ZnO through the incorporation of Al can significantly improve the photovoltage and PEC activity of the photocathode, leading to a record-high photovoltage of ~0.53 V_{RHE}. By capping the photocathode with a crystal-engineered TiO_2 protective layer, we are able to significantly stabilize the photocathode against photocorrosion and further improve the PEC activity of the final p-(CuO/CuO:Al)/n-ZnO:Al/TiO_2/Au–Pd photocathode, resulting in record-high photocurrent density of ~5.4 mA cm^{-2} and photocorrosion stability of ~87% after 5 hours.

Introduction

Photoelectrochemical (PEC) water splitting has emerged as a potential technology for converting solar energy into storable hydrogen fuel with low environmental impact.[1–10] Copper oxide is a promising photocathode material for PEC water splitting applications because of its high optical absorption, favorable band alignment for water reduction, inherently p-type character, low fabrication cost, and abundance in the Earth's crust.[10–21] Cuprous oxide (Cu_2O) and cupric oxide (CuO) are the two most common forms of copper oxide. Cu_2O has an optical bandgap of around 2.2 eV and a theoretically predicted photocurrent of 14.7 mA cm^{-2} under standard air mass 1.5 global (AM 1.5G) irradiation.[22] CuO, on the other hand, has an optical bandgap of around 1.5 eV that is ideal for solar energy harvesting and a theoretically predicted photocurrent density of 35 mA cm^{-2} under standard illumination of AM 1.5 G.[22,23] These characteristics make CuO an excellent candidate for water splitting driven by visible light. However, only a handful of reports in the literature have investigated and developed CuO photocathodes for PEC water splitting applications and hydrogen evolution.[24–31]

One of the main limiting factors for using CuO in a PEC water splitting system is the poor photocorrosion stability of CuO photocathodes in aqueous solution. To practically use CuO-based photocathodes for commercial applications, the issue of long-term photocorrosion stability has to be resolved. An efficient method to improve the photocorrosion stability of photoelectrodes in electrolyte solution is to cover them with a protective layer.[32–38] An example of such a layer for large-scale deployment is sputter-deposited titanium dioxide (TiO_2).[35,36] However, while a TiO_2 protective layer improves photocorrosion stability over a short duration of less than one hour, it has so far performed poorly in long-term photocorrosion stability tests of several hours.[8,17] Therefore, for practical applications, there is a need to engineer the TiO_2 protective layer to suppress photocorrosion over a much longer duration.

It is also of great interest to alter the material properties of CuO itself to stabilize it against photocorrosion. Recently, we

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† Electronic supplementary information (ESI) available. See DOI: 10.1039/c8ta03027h

have shown that the photocatalytic activity of a CuO-based photocathode can be improved by crystal engineering and tuning the chemical composition of the thin CuO film.[39–41] However, these methods have limited impact on the long-term photocorrosion stability of CuO-based photocathodes. Therefore, there is a need for further development to improve the material properties of CuO for higher photocurrent and better photocorrosion stability.

Another major limiting factor of CuO photocathodes is the relatively low photovoltage, which necessitates a large external bias to carry out water splitting and therefore restricts the energy conversion efficiency and hydrogen generation when such photocathodes are used in a PEC cell. The main reason for a photocathode's low photovoltage is an interface energetic mismatch between the semiconductor and the electrolyte.[42,43] Forming a buried junction by incorporating an ohmic-like semiconductor has emerged as a promising approach to decouple the photovoltage from the electrolyte redox potential and generate the photovoltage from an internal junction, thereby improving the onset potential for hydrogen evaluation reaction (HER).[44–46] Most recently, Septina *et al.*[28] demonstrated that inserting a thin layer of cadmium sulfide (CdS) semiconductor between the CuO and the TiO_2 protective layer can increase the photovoltage to 0.45 V, although their method improved the photocorrosion stability only marginally. Kargar *et al.*[26] have also shown that the insertion of a thin semi-insulating film of n-type zinc oxide (ZnO) between CuO and TiO_2 can improve the H_2 generation. However, this method does not improve the photovoltage. Therefore, to enhance the photovoltage and improve the PEC activity of CuO photocathodes, a highly conductive buried n-type semiconductor needs to be incorporated and optimized.

In this paper, we demonstrate that incorporating Al into a thin CuO film can significantly improve its photocorrosion stability. Al-incorporated CuO (CuO:Al) photocathode is prepared by co-sputtering CuO and Al onto a glass substrate coated with fluorine-doped tin oxide (FTO). By controlling the amount of Al incorporated into the thin CuO film and inserting a thin CuO interfacial layer between the FTO-coated substrate and the CuO:Al, photocorrosion stability of 92% over 1 hour is achieved, along with a short-circuit current J_{sc} of 3.7 mA cm^{-2}, which are among the highest reported values for plain CuO photocathodes without co-catalyst and protective layer. We also show that a highly conductive thin film of n-type Al-incorporated ZnO (ZnO:Al) on top of the thin CuO:Al film, forming a buried heterojunction with p-(CuO/CuO:Al) on one side and ohmic-like contact with the electrolyte on the other side, results in significant enhancement of the photovoltage and photocurrent, with the photovoltage reaching a record-high value of ~0.53 V *vs.* RHE (V_{RHE}). Furthermore, we demonstrate that nanocrystal engineering of the TiO_2 protective layer through control of the fabrication parameters is an effective method to improve its protective efficacy. By covering the p-(CuO/CuO:Al)/n-ZnO:Al photocathode with this nanocrystal-engineered TiO_2 protective layer and depositing Au–Pd nanoparticles, record-high photocurrent density of ~5.4 mA cm^{-2} and photocorrosion stability of ~87% over 5 hours were achieved.

Results and discussion

Al-incorporated CuO (CuO:Al) photocathode

Cross-sectional transmission electron microscopy (TEM) images of CuO and CuO:Al films are shown in Fig. S1,† with the CuO film (without Al) exhibiting columnar growth. The amount of Al incorporated into the thin CuO film, which is controlled by the Al sputtering power, is an important factor for tuning the PEC activity of CuO:Al photocathodes. The bulk and surface elemental compositions of the CuO:Al films were obtained using EDS (energy dispersive X-ray spectroscopy) and XPS, and the extracted values are listed in Tables S1 and S2, respectively, of the ESI† to provide more information about the chemical composition of the sputter-deposited thin CuO:Al films.

Fig. 1(a) plots the photocurrent density (J_{photo}) and dark current density (J_{dark}) *vs.* voltage for CuO:Al photocathodes fabricated with Al sputtering power of 0–20 W. The photovoltage increases slightly for higher Al sputtering power, reaching ~0.41 V_{RHE} for Al sputtered at 12 W. Although the photocurrent improves with increasing Al incorporation, the dark current also worsens due to an increase in recombination centers. To determine the optimum trade-off between J_{photo} and J_{dark} of the CuO:Al photocathodes, (J_{photo}–J_{dark}) is plotted in Fig. 1(b). As the Al sputtering power increases from 0 W to 12 W, the value of (J_{photo}–J_{dark}) increases and reaches a maximum magnitude of 2.85 mA cm^{-2} at 0 V_{RHE}. Thereafter, (J_{photo}–J_{dark}) decreases at higher Al sputtering power. The initial improvement in PEC performance of CuO:Al photocathodes with increasing Al sputtering power is mainly due to enhancement of optical absorption (see Fig. S2 of ESI†), while the subsequent performance degradation at Al sputtering power above 12 W is likely caused by an increase in recombination rate and a reduction in electron generation.

A high dark current in CuO:Al photocathodes could also be due to the occurrence of direct contact between Al nanoparticles and the conductive FTO electrode, shunting the photocathode and increasing charge carrier recombination. To investigate this hypothesis and overcome the problem, a thin CuO interfacial layer was inserted between the FTO-coated glass substrate and the CuO:Al film. Fig. S3 in the ESI† shows a cross-sectional high-resolution TEM (HR-TEM) image of a fabricated p-(CuO/CuO:Al) photocathode with the CuO interfacial layer having a thickness of ~30 nm. Fig. 2 shows that the CuO interfacial layer in a p-(CuO/CuO:Al) photocathode does not appear to affect the photocurrent, but reduces the dark current significantly. This highlights the importance of the CuO interfacial layer in reducing the interfacial recombination losses.

To investigate the photocorrosion stability of p-CuO/CuO:Al photocathodes prepared using Al sputtering power of 1–20 W, the current density of the photocathodes was measured under continuous illumination for two hours. The long-term photocurrent measurements in Fig. 3 show that the incorporation of Al significantly improves the photocorrosion stability of the photocathodes. p-CuO/CuO:Al photocathodes prepared using

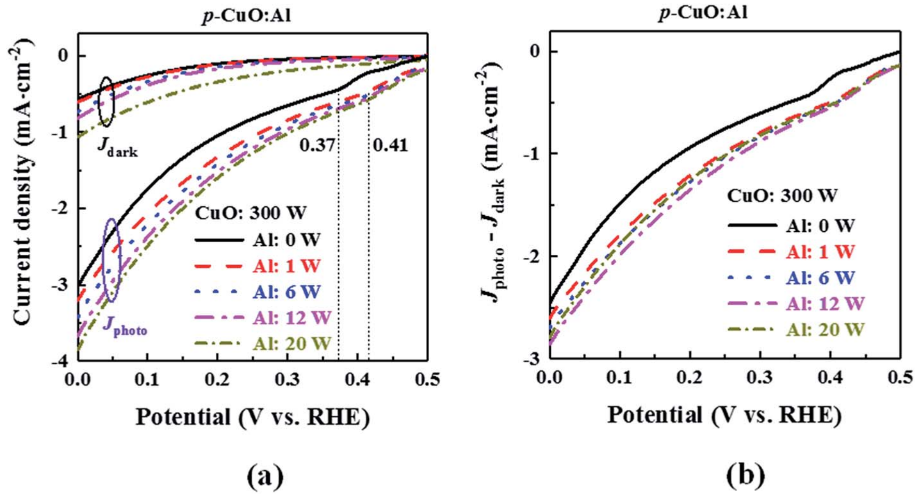

Fig. 1 (a) Current density and (b) (J_{photo}–J_{dark}) *vs.* potential for CuO:Al photocathodes fabricated with Al sputtering power of 0–20 W. Although the photocurrent improves with increasing Al incorporation, the dark current also worsens.

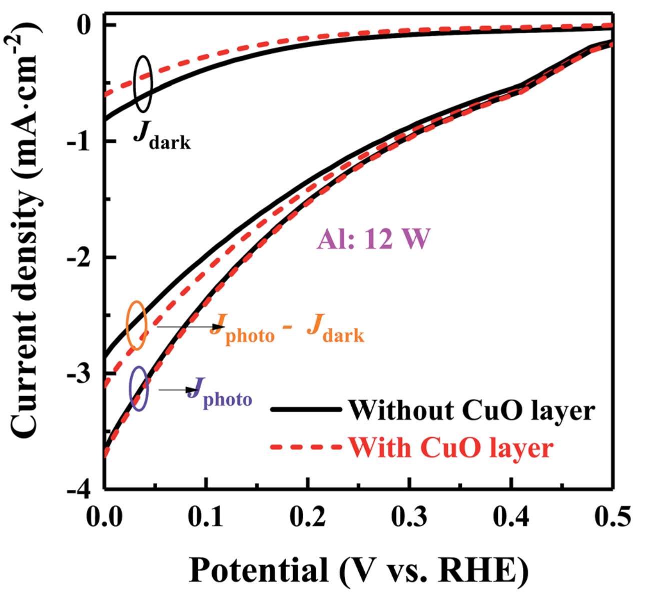

Fig. 2 J_{photo}, J_{dark}, and (J_{photo}–J_{dark}) *vs.* potential for CuO:Al and CuO/CuO:Al photocathodes fabricated at Al sputtering power of 12 W. Inserting a CuO interfacial layer reduces the dark current significantly without affecting the photocurrent.

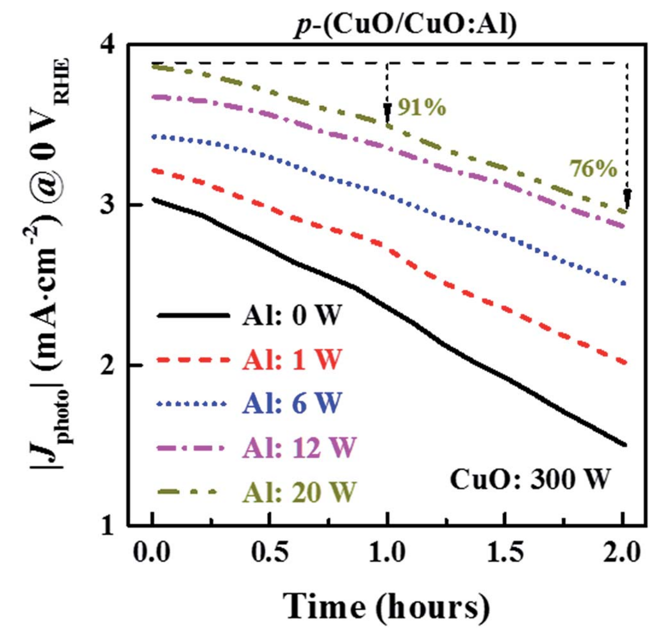

Fig. 3 Photocorrosion stability of p-(CuO/CuO:Al) photocathodes fabricated with Al sputtering power of 0–20 W. Photocathodes prepared using Al sputtering power of 0, 1, 6, 12, and 20 W retained 49%, 61%, 63%, 78%, and 76% of their initial J_{photo} at 0 V_{RHE}, respectively, after 2 hours. This indicates the effectiveness of Al incorporation in improving the photocorrosion stability.

Al sputtering power of 1, 6, 12, and 20 W retained 85%, 89%, 92%, and 91% of their initial J_{photo} at 0 V_{RHE}, respectively, after 1 hour. On the other hand, the control photocathode (CuO without Al) achieved only 77% photocorrosion stability over the same duration. The best-performing p-CuO/CuO:Al photocathode (Al sputtering power of 12 W) shows a photocorrosion stability of 78% after 2 hours of continuous illumination, which is around 1.6 times that of the control photocathode (49%). As the Fermi level of CuO is not energetic enough to rapidly transfer the photogenerated electrons to the electrolyte solution, the accumulation of electrons at the surface causes the CuO to be reduced to Cu_2O, resulting in photocorrosion of the CuO photocathode. This reduction of CuO to Cu_2O decreases optical absorption in the visible light regime, leading to inferior photocathode performance. With the incorporation of Al in CuO, the reduction of CuO to Cu_2O is suppressed by the facilitated transfer of photogenerated electrons from the surface of the photocathode to the electrolyte, thereby improving the photocorrosion stability.

To further study the photocorrosion stability of the CuO and CuO:Al photocathodes, X-ray photoelectron spectroscopy (XPS),

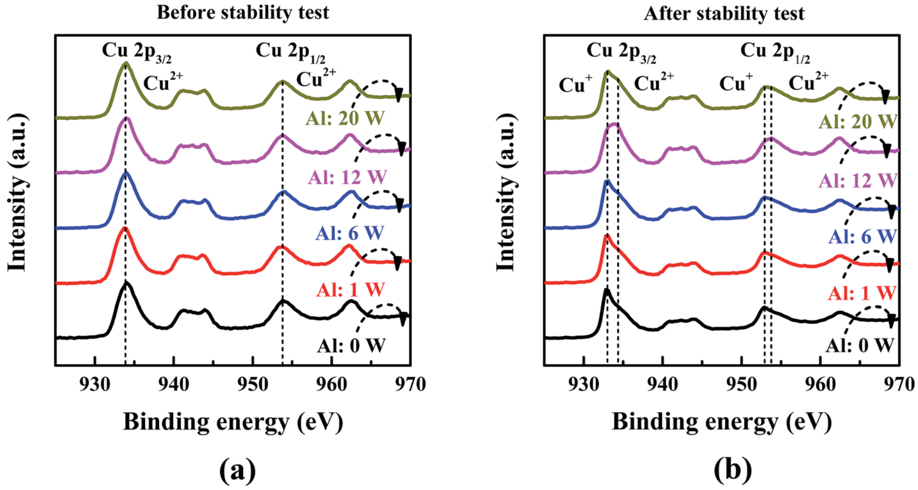

Fig. 4 Cu 2p XPS spectra of CuO:Al photocathodes (a) before and (b) after photocorrosion stability test. Small peaks at binding energies of 932.7 and 952.8 eV, which are characteristic of Cu_2O, appear for both CuO and CuO:Al photocathodes after photocorrosion stability test, but these peaks are weaker in the CuO:Al photocathodes.

X-ray diffraction (XRD), and scanning electron microscopy (SEM) analyses were carried out on the same photocathodes before and after the photocorrosion stability test. It can be seen from Fig. 4(a) that the XPS spectra of the CuO and CuO:Al photocathodes before the stability test show two main peaks at binding energies of 933.8 and 954 eV, which correspond to the $2p_{3/2}$ and $2p_{1/2}$ peaks of Cu^{2+}, respectively, indicating that CuO is the dominant phase.[47,48] Satellite peaks at higher binding energies of around 940.8, 943.8, and 962.5 eV further confirm CuO as the dominant phase.[49] The XPS spectra of the photocathodes after the photocorrosion stability test are shown in Fig. 4(b), revealing the emergence of small peaks at binding energies of 932.7 and 952.8 eV for both CuO and CuO:Al photocathodes. These are the $2p_{3/2}$ and $2p_{1/2}$ peaks of Cu^+, respectively, and are characteristic of Cu_2O.[50,51] The presence of the Cu^+ peaks is confirmed by deconvolution of the XPS spectra. Fig. 5(a) and (b) show, respectively, the deconvoluted Cu $2p_{3/2}$ peaks of the control sample (CuO) and the CuO:Al photocathode with Al sputter power of 12 W. The ratio of the areas under the Cu^+ and Cu^{2+} $2p_{3/2}$ peaks ($A_{Cu^+}/A_{Cu^{2+}}$) is calculated for each of the photocathodes and shown in Fig. 5(c). This ratio is significantly lower for the CuO:Al photocathode with Al sputtered at 12 W ($A_{Cu^+}/A_{Cu^{2+}} = 0.122$) than for the control CuO photocathode ($A_{Cu^+}/A_{Cu^{2+}} = 0.485$), indicating that the incorporation of Al in CuO curbs the reduction of CuO and helps to improve its photocorrosion stability.

The XRD spectra of CuO and CuO:Al photocathodes before and after the photocorrosion stability test are shown in Fig. S4.† The XRD spectrum corresponding to the FTO was removed to clearly illustrate the XRD spectra of the CuO and CuO:Al films. Before the stability test, all samples show XRD peaks at 2θ of 35.482 and 38.764 degrees, which are ascribed to the (002) and (111) planes of CuO (JCPDS# 05-0661), respectively. An additional XRD peak at 2θ of 36.45 degrees, attributed to the (111) plane of Cu_2O (JCPDS# 05-0667), is observed for all samples after the stability test. It can be seen that the intensity of the Cu_2O XRD peaks is significantly reduced by increasing the Al content of CuO:Al, indicating less reduction of CuO to Cu_2O in CuO:Al photocathodes compared to the control CuO photocathode.

SEM images of the CuO and CuO:Al photocathodes (Fig. S5†) show the formation of small islands on the surface of the samples after the photocorrosion stability test, which is mainly caused by the reduction of CuO to Cu_2O.[52] The distribution of the islands is significantly influenced by Al incorporation during the deposition of CuO:Al. Compared to the control CuO sample, the formation of islands is reduced for CuO:Al photocathodes, providing further evidence of the improvement in photocorrosion stability with the incorporation of Al in CuO.

The PEC activity of the CuO and CuO:Al photocathodes is analyzed using electrochemical impedance spectroscopy (EIS) to determine the charge transfer rate at the CuO/electrolyte interface and obtain the flat-band voltage and majority carrier density in the CuO. As shown in Fig. 6(a), the Nyquist plot for the photocathodes at a potential of 0 V_{RHE} under illumination exhibits semicircle characteristics at low frequencies, which indicates a charge transfer process at the interface between the semiconductor and the electrolyte.[53] Charge transfer resistance (R_{ct}) is represented by the semicircle radius. It can be seen from Fig. 6(a) that R_{ct} is significantly reduced when Al is incorporated in the CuO, with R_{ct} decreasing with increasing Al sputtering power up to 12 W. This variation of R_{ct} with Al sputtering power is in good agreement with the photocurrent density. Thus, the incorporation of an optimum amount of Al in the CuO facilitates the transfer of photo-induced electrons across the semiconductor/electrolyte interface and results in better PEC performance.

Fig. 6(b) shows the linear part of the Mott–Schottky (M–S) plot for the photocathodes, with a negative slope that is indicative of the p-type nature of the thin CuO and CuO:Al films. This linear part can be fitted using the following M–S equation,

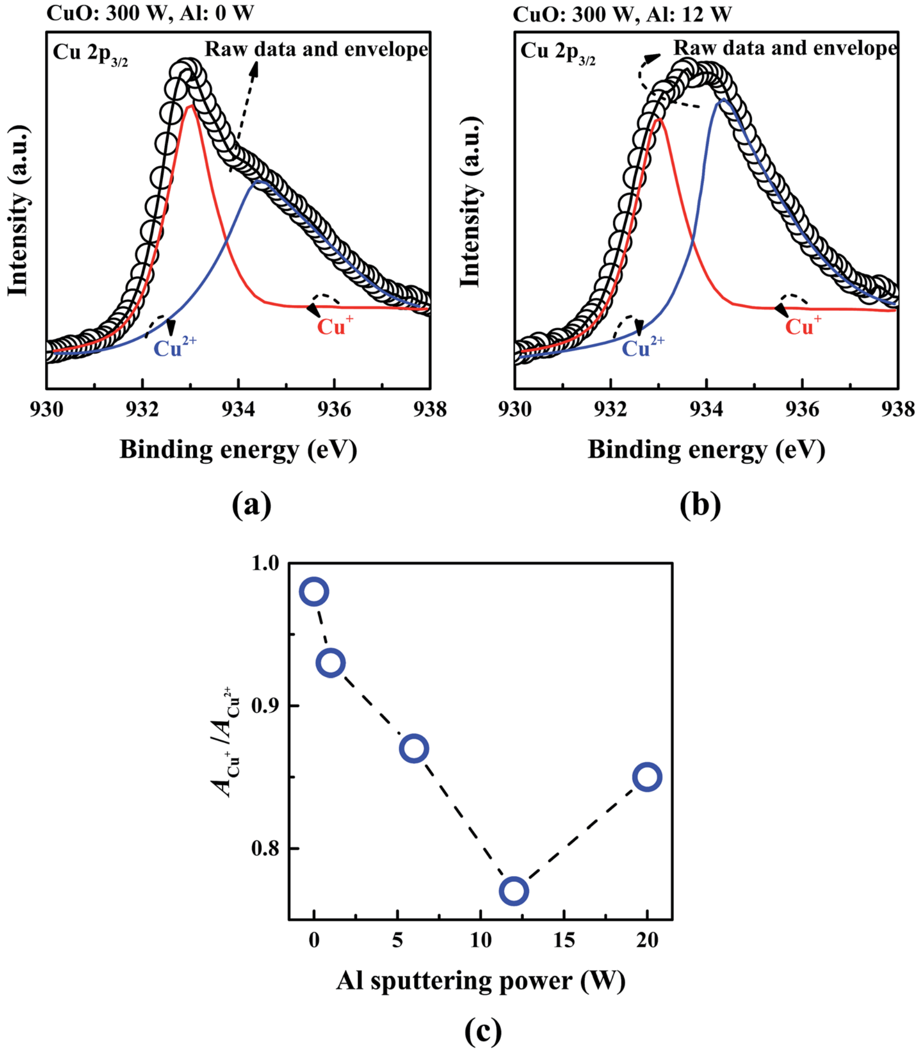

Fig. 5 Experimentally measured Cu $2p_{3/2}$ XPS spectra (circles) and fitted curves (lines) for the (a) CuO and (b) CuO:Al photocathodes after photocorrosion stability test. The Cu^+ peak of Cu_2O is weaker in the CuO:Al photocathode. The ratio of the area of the Cu^+ peak to the area of the Cu^{2+} peak ($A_{Cu^+}/A_{Cu^{2+}}$) is significantly influenced by Al concentration in the CuO, as shown in (c).

which provides a qualitative comparison for samples with similar device geometry and material composition:[54]

$$\frac{1}{C^2} = \frac{2}{N_A e \varepsilon \varepsilon_0}\left[(V_S - V_{FB}) - \frac{kT}{e}\right] \tag{1}$$

where C is the space charge capacitance in the semiconductor, N_A is the hole carrier density, e is the elemental charge, ε is the relative permittivity of the semiconductor (10.26 for CuO), ε_0 is the permittivity of vacuum, V_S is the potential applied to the photocathode, V_{FB} is the flat-band voltage, k is the Boltzmann constant, and T is the temperature. The values of N_A and V_{FB} are thus estimated from the slope and the x-intercept, respectively, of the extrapolated linear part of the M–S curve. Fig. 6(c) and (d) plot the extracted values of N_A and V_{FB}. The variation of N_A and V_{FB} with Al sputtering power also agrees well with the photocurrent. The CuO:Al with Al sputtered at 12 W has N_A of 7.5×10^{21} cm^{-3}, which is ~1 order of magnitude higher than that of the control CuO film, signifying faster charge transfer and hence improved photocatalytic activity. The V_{FB} values of the CuO:Al photocathodes are higher than that of the control CuO sample, providing a larger space charge region potential and therefore a larger driving force to separate the photo-induced electron–hole pairs in the space charge region, which in turn also results in improved PEC performance.

p-(CuO/CuO:Al)/n-ZnO:Al photocathode

Although the photovoltage of the CuO:Al photocathodes is slightly enhanced by the incorporation of Al into the thin CuO film, it is still far from the maximum achievable photovoltage (MAP). The low photovoltage of the CuO:Al photocathodes is mainly attributed to the poor interface between the CuO:Al and the electrolyte.[42,43] The MAP is determined by the degree of band bending,[44] so to increase the photovoltage, it is necessary to shift the positions of the band edges. Forming a buried junction is an efficient method for this purpose.[45] A metal with a suitable work function or a thin film of heavily doped n-type

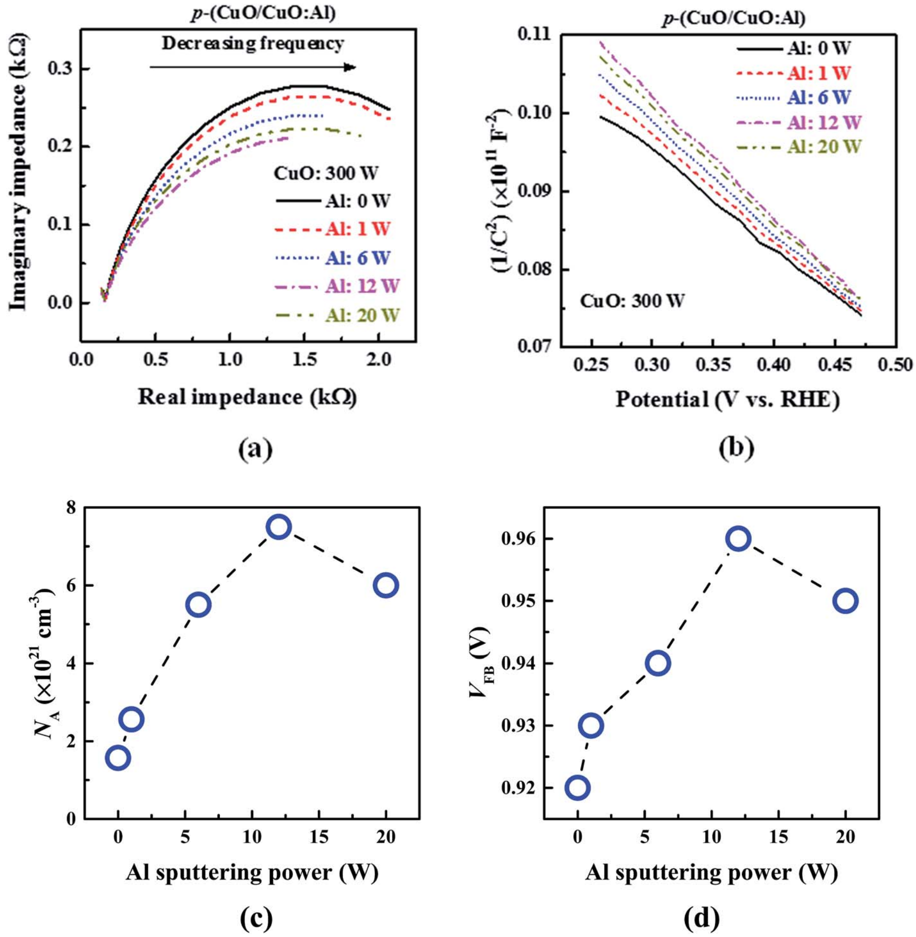

Fig. 6 (a) Nyquist plot, (b) Mott–Schottky plot, (c) hole carrier density, and (d) flat band voltage of CuO:Al photocathodes fabricated with Al sputtering power of 0–20 W. Charge transfer resistance is significantly reduced while hole carrier density and flat band potential are considerably increased with Al incorporation.

semiconductor with sufficiently negative conduction and valence band edge energies is required to form a buried junction and increase the band bending in p-type CuO:Al. A suitable candidate is Al-incorporated ZnO (ZnO:Al), which has a conduction band minimum at −4.6 eV and valence band maximum at −8.07 eV.[55,56] In addition, one advantage of ZnO over the CdS used by Septina *et al.*[28] is that Zn is more abundant than Cd.

p-(CuO/CuO:Al)/n-ZnO:Al photocathodes were fabricated with Al sputtering power of 0–30 W used for ZnO:Al deposition. Fig. S6† shows a cross-sectional TEM image of one such photocathode with Al sputtered at 20 W during ZnO:Al deposition, as well as an HR-TEM image zoomed in on the ZnO:Al layer. The Al sputtering power determines the Al content in the thin ZnO:Al film, which is a key factor influencing the doping concentration and conductivity of the ZnO:Al and can therefore affect the PEC performance of the photocathodes. From Fig. 7(a) and (b), it can be seen that the insertion of the thin ZnO:Al film between the electrolyte and the CuO:Al semiconductor improves the photovoltage and the photocurrent, achieving a record-high photovoltage ~0.53 V_{RHE} for the photocathode with Al sputtered at 20 W during ZnO deposition. This is mainly attributed to the reduction of the CuO:Al/electrolyte interface energetics mismatch and alleviation of the voltage requirement from the photoanode, resulting in improved water splitting efficiency. Fig. 7(c) shows that the photocorrosion stability of the photocathodes is also significantly improved by the ZnO:Al layer, as the electrostatic field at the p-(CuO/CuO:Al)/n-ZnO:Al heterojunction improves the extraction of photogenerated carriers. For both photocurrent and photocorrosion stability, higher Al sputtering power up to 20 W during ZnO:Al deposition gives better performance, while further increasing the Al content reduces the PEC activity of the photocathode.

To investigate the origin of the effects of ZnO Al concentration on the PEC activity of p-(CuO/CuO:Al)/n-ZnO:Al photocathodes, EIS measurements were performed under illumination. Fig. 8(a) shows the resulting Nyquist plot. The semicircle radius is reduced with increasing Al content in the thin ZnO:Al film, indicating more efficient separation of the photogenerated electron–hole pairs and smaller resistance for interfacial charge transfer. However, increasing the Al sputtering power above 20 W during ZnO:Al deposition increases the charge transfer resistance R_{ct} (larger semicircle radius). This

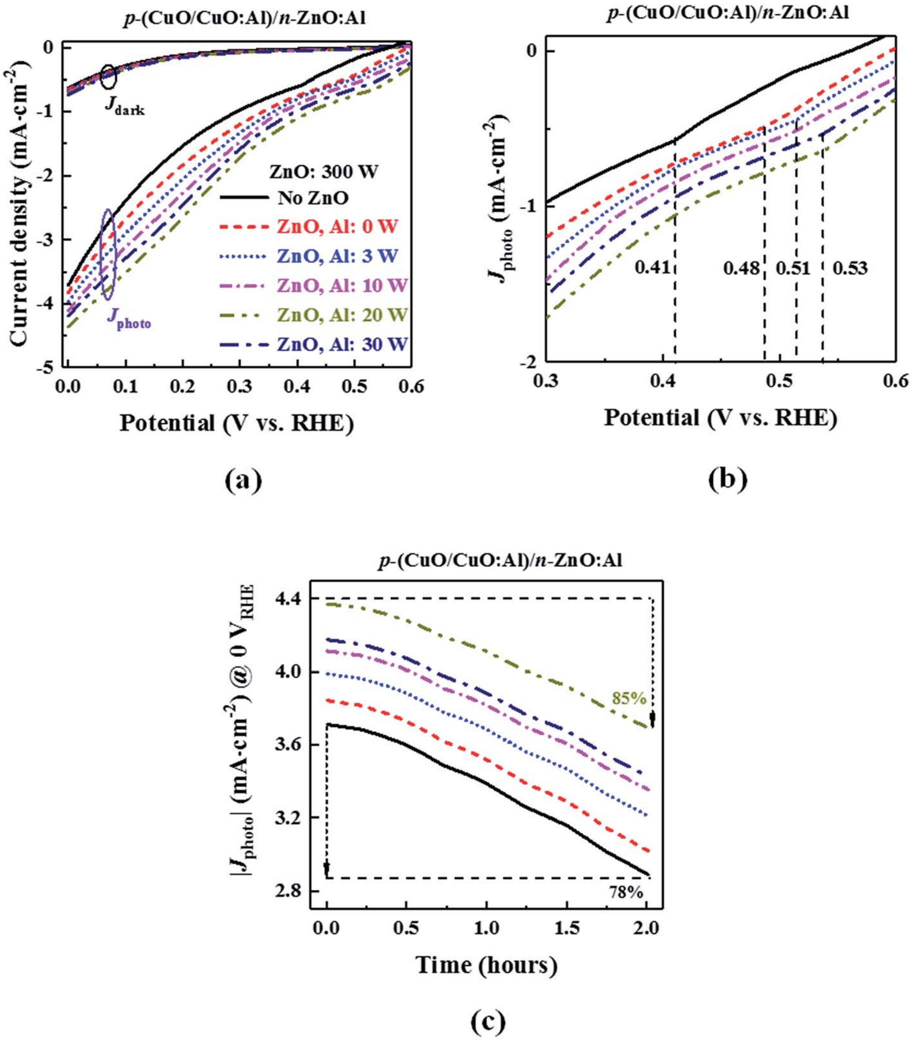

Fig. 7 (a) Current density, (b) zoomed-out portion of the photocurrent, and (c) photocorrosion stability of the p-(CuO/CuO:Al)/n-ZnO:Al photocathodes. Both plots share the same legend. Insertion of a thin ZnO:Al film between the electrolyte and the CuO:Al semiconductor improves the photovoltage, photocurrent, and photocorrosion stability. The photocathode with Al sputtered at 20 W during ZnO deposition achieved a record-high photovoltage of ~0.53 V_{RHE} and retained 85% of its initial J_{photo} at 0 V_{RHE} after 2 hours.

trend shows good correlation with the measured PEC activity in Fig. 7(a). The exchange current density and kinetic energy required to overcome the energy barrier in a faradaic reaction are the two key factors that impact the charge transfer resistance.[57] Therefore, the smaller charge transfer resistance of the p-(CuO/CuO:Al)/n-ZnO:Al photocathodes compared to that of the control p-(CuO/CuO:Al) photocathode implies promotion of the water-splitting reaction, leading to faster reduction of H^+ to H_2 and improving the hydrogen evolution reaction.

The conduction band edge of ZnO:Al is lower than that of CuO:Al, allowing photogenerated electrons to flow easily from the CuO:Al to the ZnO:Al. Therefore, this conducive band alignment of the p-(CuO/CuO:Al)/n-ZnO:Al heterojunction supports the efficient separation of photogenerated electron–hole pairs and improves the harvesting of light. To gain more insights into the effect of the ZnO:Al layer on the band bending in the CuO:Al, M–S measurements were carried out on the p-(CuO/CuO:Al)/n-ZnO:Al photocathodes under illumination. From the M–S plot in Fig. 8(b), the variation in V_{FB} (x-intercept) with the Al content of ZnO:Al shows that the ZnO:Al film with Al sputtered at 20 W has the highest impact on V_{FB}. The downward band bending in the semiconductor (V_B) is given by $V_B = -(V_S - V_{FB})$. Since the potential V_S applied to the photocathode is negative, a more positive value of V_{FB} through incorporation of an optimized layer of ZnO:Al increases V_B. More band bending in the photocathode at the interface with the electrolyte reduces the charge transfer resistance, facilitates charge separation, and suppresses charge recombination, leading to enhancements in photovoltage and photocurrent as seen in Fig. 7(a).

Hall measurements were also performed on a series of thin ZnO:Al films deposited on glass substrate using Al sputtering power of 1–30 W. The electrical resistivity and hall mobility of these ZnO:Al films on glass substrate are shown in Fig. S7 of the ESI.† It can be seen that by increasing the Al sputtering power from 1 W to 20 W, the electrical resistivity of the ZnO:Al reduces and its hall mobility increases. This improvement in electrical

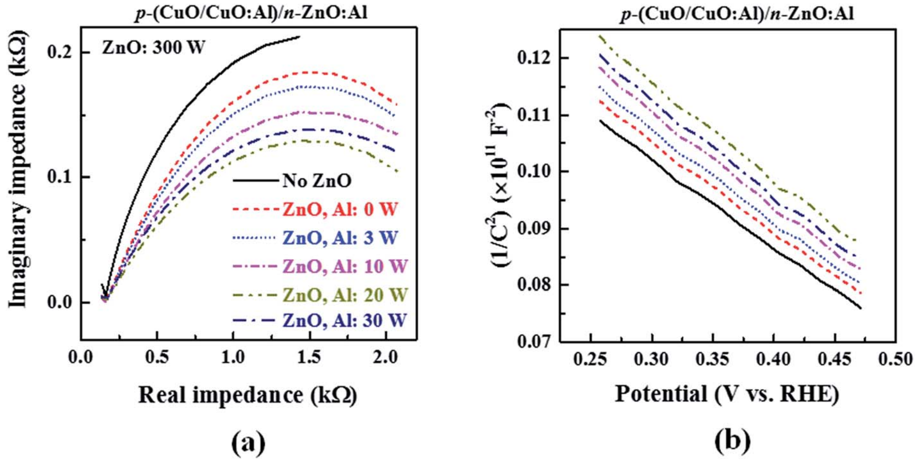

Fig. 8 (a) Nyquist and (b) Mott–Schottky plots of the p-(CuO/CuO:Al)/n-ZnO:Al photocathodes. Both plots share the same legend. Charge transfer resistance and hole carrier concentration are significantly improved with the insertion of a thin ZnO:Al film between the electrolyte and the CuO:Al semiconductor.

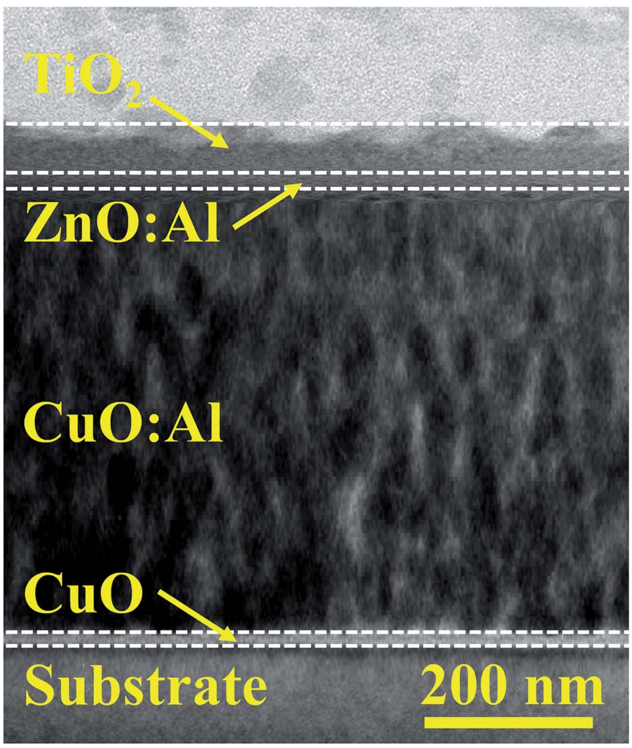

Fig. 9 Cross-sectional TEM image of a p-(CuO/CuO:Al)/n-ZnO:Al/TiO_2 photocathode with TiO_2 sputtered at 200 W. The thickness of the TiO_2 protective layer is ~20 nm.

characteristics of the thin ZnO:Al film at higher Al content is mainly attributed to an increase in Al interstitials and enhancement of substitutional Al^{3+} ions. However, when the Al sputtering power is increased further to 30 W, the electrical resistivity of the ZnO:Al film increases due to segregation of electrically inactive dopants into the grain boundaries.[58] The electrical characteristics of the ZnO directly affect R_{ct}, and are thus correlated with the PEC activity of the photocathodes.

(p-(CuO/CuO:Al)/n-ZnO:Al)/TiO_2/Au–Pd photocathode

Little research has been conducted to study the impact of the material properties of a TiO_2 protective layer on the long-term photocorrosion stability of photocathodes. Nanocrystal engineering by controlling the fabrication parameters is a possible method to tune the crystal quality and material properties of thin metal oxide films. A 20 nm-thick film of TiO_2 was thus deposited on p-(CuO/CuO:Al)/n-ZnO:Al photocathodes by sputtering at various RF powers ranging from 50 to 200 W. The sputtering power of the TiO_2 significantly influences the thickness of the deposited TiO_2 thin film. To keep the thickness of the sputtered TiO_2 constant, we tuned the deposition time. Fig. 9 shows a cross-sectional TEM image of a p-(CuO/CuO:Al)/n-ZnO:Al/TiO_2 photocathode with TiO_2 sputtered at 200 W.

Au–Pd nanoparticles were deposited on top of the TiO_2 by sputtering from a Au–Pd (60 : 40) target using the same conditions as in our recent work,[39] where these nanoparticles were demonstrated to be efficient co-catalysts for improving the PEC activity of CuO photocathodes. The size and distribution of the Au–Pd nanoparticles play an important role in the PEC activity of CuO-based photocathodes. An optimal distribution of Au–Pd nanoparticles was obtained using the method presented in our earlier work.[39] The SEM image in Fig. S8 of the ESI† shows Au–Pd nanoparticles randomly and uniformly distributed on the TiO_2 protective layer.

Fig. 10(a) shows the effect of TiO_2 sputtering power on the long-term photocorrosion stability of p-(CuO/CuO:Al)/n-ZnO:Al/TiO_2/Au–Pd photocathodes. Increasing the TiO_2 sputtering power from 50 W to 200 W significantly improves the photocorrosion stability, which can be attributed to an improvement in the crystallinity of the TiO_2 layer. Even after 5 hours of illumination, the photocathode with TiO_2 sputtered at 200 W maintained a photocorrosion stability of ~87%.

The photocorrosion stability of p-(CuO/CuO:Al), p-(CuO/CuO:Al)/n-ZnO:Al, and p-(CuO/CuO:Al)/n-ZnO:Al/TiO_2/Au–Pd photocathodes are compared in Fig. 10(b). After 2 hours of

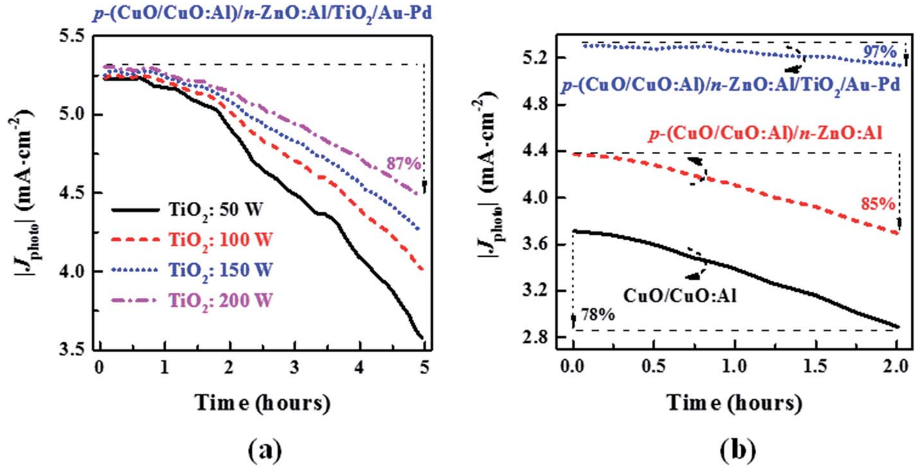

Fig. 10 (a) Photocorrosion stability of p-(CuO/CuO:Al)/n-ZnO:Al/TiO_2/Au–Pd photocathodes fabricated with TiO_2 sputtering power from 50 W to 200 W. After 5 hours of illumination, the photocathode with TiO_2 sputtered at 200 W maintained a photocorrosion stability of ~87%. (b) Comparison of the photocorrosion stability of p-(CuO/CuO:Al), p-(CuO/CuO:Al)/n-ZnO:Al, and p-(CuO/CuO:Al)/n-ZnO:Al/TiO_2/Au–Pd photocathodes.

illumination, the p-(CuO/CuO:Al)/n-ZnO:Al/TiO_2/Au–Pd photocathode (TiO_2 sputtered at 200 W) shows significantly enhanced photocorrosion stability of 97% compared with the p-(CuO/CuO:Al) and p-(CuO/CuO:Al)/n-ZnO:Al photocathodes, which achieved photocorrosion stability of 78% and 85% respectively. This demonstrates the protective role of the TiO_2 layer. The small but still-present decrease in photocurrent of the p-(CuO/CuO:Al)/n-ZnO:Al/TiO_2/Au–Pd photocathode with increasing illumination time could be because the TiO_2 layer deposited at 200 W is still slightly amorphous.[17]

The influence of the TiO_2 sputtering power on the crystal quality of the thin TiO_2 film was investigated using XRD. Fig. 11(a) shows the XRD spectrum for each of the TiO_2 films deposited at the various sputtering powers. All the TiO_2 films show XRD peaks at 2θ of 25.28 and 37.8 degrees, which are associated with the anatase TiO_2(101) and TiO_2(004), respectively (JCPDS 84-1286). Fig. 11(b) plots the full-width-half-maximum (FWHM) of the main TiO_2(101) peak *vs.* TiO_2 sputtering power. By increasing the TiO_2 sputtering power, the intensity of the XRD peaks is increased and the FWHM of the main TiO_2(101) peak is significantly reduced, indicating an improvement in the crystallinity of the thin TiO_2 film.

High-resolution XPS was used to study how the crystallinity of the TiO_2 film affects its surface chemical composition before and after the 5 hour photocorrosion stability test. Fig. 12(a) and (b) show the Ti 2p core-level XPS spectra of the TiO_2 protective layer before and after the stability test. The main peaks at 458.58 and 464.2 eV correspond to Ti $2p_{3/2}$ and Ti $2p_{1/2}$, respectively. The Ti $2p_{3/2}$ peaks were deconvoluted into their Ti^{4+} and Ti^{3+} components at respective binding energies of 458.6 and 457.1 eV, and the ratio of the area under the Ti^{3+} peak to the area under the Ti^{4+} peak ($A_{Ti^{3+}}/A_{Ti^{4+}}$) for each TiO_2 sputtering power is

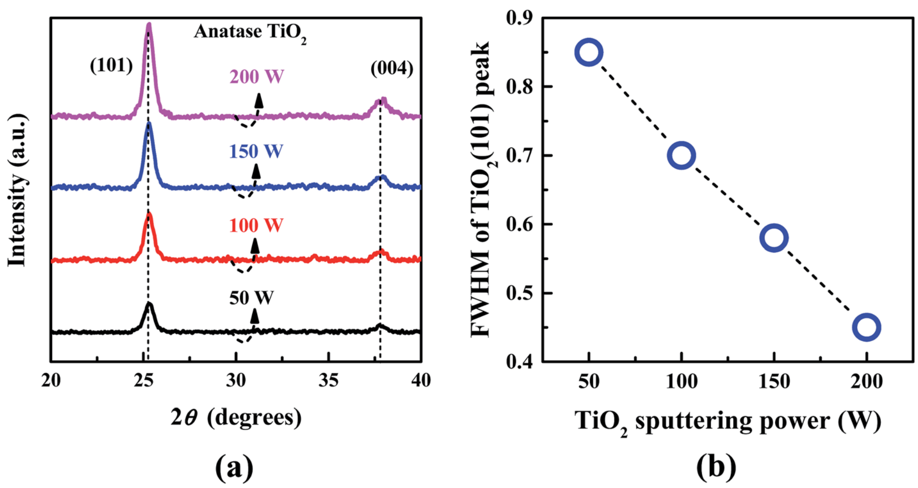

Fig. 11 (a) XRD spectrum for each of the TiO_2 films deposited at the various sputtering powers and (b) FWHM of the main TiO_2(101) peak *vs.* TiO_2 sputtering power. Higher TiO_2 sputtering power gives better crystallinity.

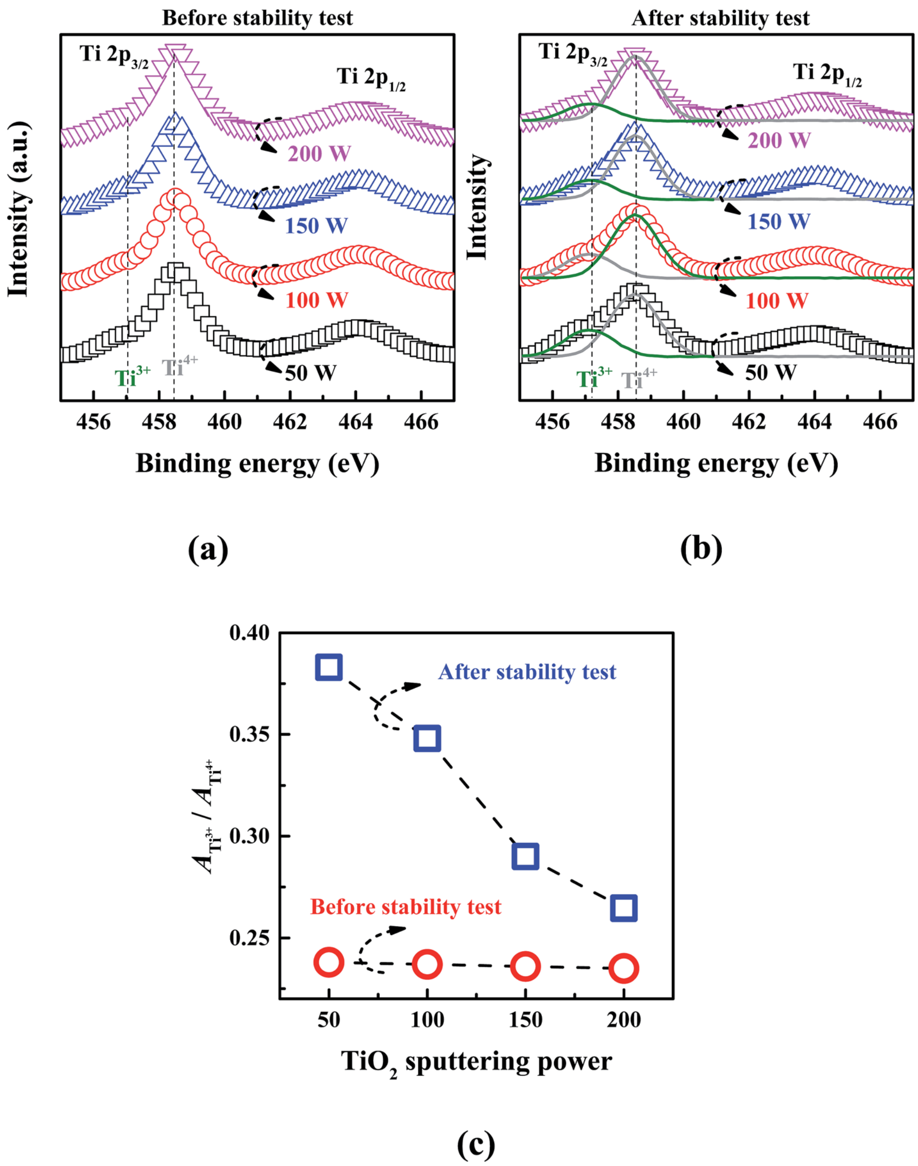

Fig. 12 High-resolution Ti 2p core-level XPS spectra of the TiO_2 protective layer (a) before and (b) after the 5 hour photocorrosion stability test. (c) The ratio of the area under the Ti^{3+} peak to the area under the Ti^{4+} peak ($A_{Ti^{3+}}/A_{Ti^{4+}}$) for each TiO_2 sputtering power before and after the photocorrosion stability test. Increasing the sputtering power significantly suppresses the formation of Ti^{3+} states under illumination.

determined from the fitted XPS spectra and presented in Fig. 12(c). It can be seen that the $A_{Ti^{3+}}/A_{Ti^{4+}}$ ratio before the stability test is only marginally influenced by TiO_2 sputtering power, indicating that the TiO_2 sputtering power does not affect the formation of Ti^{3+} trap states in as-deposited TiO_2. However, the $A_{Ti^{3+}}/A_{Ti^{4+}}$ ratio after the stability test is considerably reduced from 0.383 to 0.264 as the TiO_2 sputtering power increases from 50 W to 200 W. Therefore, the improved crystallinity of the TiO_2 protective layer at higher sputtering power (as seen from XRD) significantly suppresses the formation of Ti^{3+} states under illumination, thereby improving the photocorrosion stability.

Incident photon-to-current efficiency (IPCE) is an indication of the light-harvesting capacity of the prepared photocathodes over the measured wavelength range. To further investigate the performance of the fabricated samples, the IPCE spectra of the prepared photocathodes at 0 V *versus* RHE under standard solar spectrum (AM 1.5G 100 mW cm^{-2}) are compared in Fig. 13. The IPCE spectra for the p-(CuO/CuO:Al), p-(CuO/CuO:Al)/n-ZnO:Al, and p-(CuO/CuO:Al)/n-ZnO:Al/TiO_2/Au–Pd photocathodes show that all photocathodes were able to absorb light and separate the resulting exciton to generate a photocurrent over a wide range of wavelengths. The p-(CuO/CuO:Al)/n-ZnO:Al/TiO_2/Au–Pd photocathode shows higher IPCE than the p-(CuO/CuO:Al)/n-ZnO:Al and p-(CuO/CuO:Al) photocathodes over a broad range of wavelengths due to coupling of the narrow-bandgap CuO:Al with the wide-bandgap ZnO and TiO_2. The IPCE of the p-(CuO/CuO:Al)/n-ZnO:Al/TiO_2/Au–Pd photocathode at 0 V_{RHE} is above 20% in the 400–650 nm wavelength range. A maximum IPCE value of ~50% was achieved at ~550 nm wavelength, gradually reducing to zero at ~750 nm.

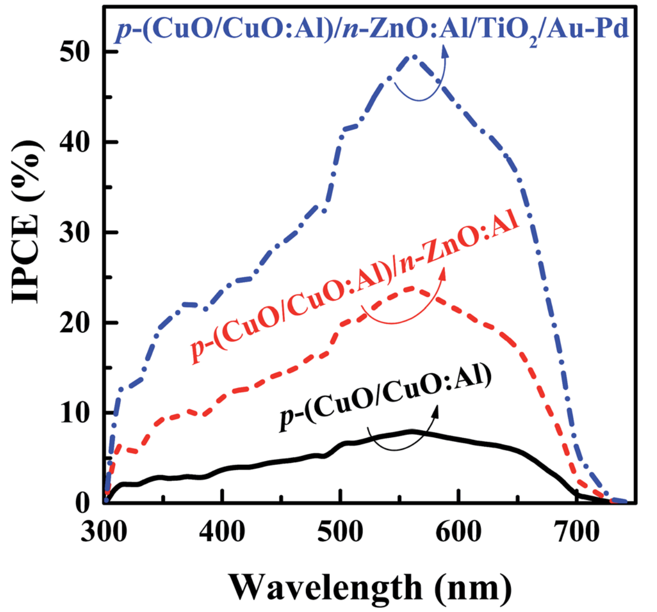

Fig. 13 IPCE spectra for the p-(CuO/CuO:Al), p-(CuO/CuO:Al)/n-ZnO:Al, and p-(CuO/CuO:Al)/n-ZnO:Al/TiO_2/Au–Pd photocathodes. Record-high J_{sc} of ~5.4 mA cm^{-2} at 0 V_{RHE} was achieved for the photocathode with TiO_2 sputtering power of 200 W. The IPCE of the p-(CuO/CuO:Al)/n-ZnO:Al/TiO_2/Au–Pd photocathode at 0 V_{RHE} is above 20% in the 400–650 nm wavelength range.

The *J–V*, hydrogen evolution, Nyquist, and M–S plots of the p-(CuO/CuO:Al)/n-ZnO:Al/TiO_2/Au–Pd photocathodes are shown in Fig. 14. As the conduction band edge of TiO_2 is lower than that of ZnO:Al, the band alignment of the p-(CuO/CuO:Al)/n-ZnO:Al/TiO_2 layer structure remains conducive for the separation of electron–hole pairs photogenerated in the CuO:Al. Improving the crystallinity of the TiO_2 protective layer facilitates the transfer of accumulated photogenerated electrons from the surface of the p-(CuO/CuO:Al)/n-ZnO:Al/TiO_2/Au–Pd photocathode to the electrolyte, resulting in significant improvement in PEC activity and hydrogen generation. Record-high J_{sc} of ~5.4 mA cm^{-2} at 0 V_{RHE} is achieved for the photocathode with TiO_2 sputtering power of 200 W. A summary of the photocurrent and photovoltage of reported CuO photocathodes for PEC water splitting applications is presented in Table 2.

The faradaic efficiency of the water-splitting reaction can be determined from the amount of hydrogen gas evolved in a given amount of time a given photocurrent, using the following equation:[59,60]

$$\text{Faradaic efficiency} = (n_{H_2})/(Q/ZF) \quad (2)$$

where n_{H_2} is the number of electrons transferred per hydrogen molecule (2 in this case), Q is the total amount of charge passed through the cell, Z is the number of moles of obtained hydrogen gas, and F is the Faraday's constant (96 485C mol^{-1}). The hydrogen evolution at an average photocurrent of 0.16 mA after 6 hours of illumination is ~60 μmol for the best-performing photocathode (TiO_2 sputtered at 200 W), giving a faradaic efficiency of ~98%.

Conclusions

In summary, we have shown that incorporating Al into CuO is an effective way to improve its photocorrosion stability. An optimum amount of Al in CuO:Al has the added benefit of enhancing the optical absorption without significantly increasing the charge recombination. We have also shown that by inserting a thin layer of CuO at the CuO:Al/FTO interface, we can prevent direct contact between the FTO and the Al nanoparticles in the CuO:Al, significantly reducing the dark current. Furthermore, the deposition of a highly conductive ZnO:Al layer on top of the CuO:Al to form a buried p-CuO:Al/n-ZnO:Al heterojunction can substantially improve the photocurrent and achieve a record-high photovoltage of ~0.53 V_{RHE}. Finally, we have demonstrated that covering the p-(CuO/CuO:Al)/n-ZnO:Al photocathode with a nanocrystal-engineered TiO_2 protective layer facilitates charge transfer from the photocathode to the electrolyte and further improves the PEC activity and photocorrosion stability of the photocathode, resulting in record-high J_{sc} of ~5.4 mA cm^{-2}, photocorrosion stability of ~87% over 5 hours, and H_2 evolution of ~60 μmol in 6 hours at an average photocurrent of 0.16 mA.

Methods

FTO-coated borosilicate float glass was used as the substrate. All substrates were sequentially cleaned by ultrasonicating in isopropyl alcohol and deionized water for ~5 minutes. The substrates were then loaded into a magnetron sputtering chamber to deposit CuO and CuO:Al. Rapid thermal annealing was performed at 550 °C for 1 minute before depositing ZnO:Al and TiO_2.

Stoichiometric sputter targets were used for the deposition of CuO, ZnO, and TiO_2, while the incorporation of Al in CuO and ZnO (to form CuO:Al and ZnO:Al, respectively) was done by co-sputtering an Al target with 99.99% purity. Radio frequency (RF) power was employed for the sputtering of all metal oxides (CuO, ZnO, and TiO_2), while direct current (DC) power was used to sputter Al. The thicknesses and sputtering powers used for the various layers are listed in Table 1 unless otherwise stated in the text. The thickness of CuO:Al was chosen to be 500 nm as this is the optimum thickness of CuO for photocatalytic applications.[20,33,34] A low sputtering power of 25 W was used to deposit the CuO interfacial layer in order to reduce interfacial damage.

For all measurements, a 0.1 M Na_2SO_4 electrolyte solution with a pH value of 5.84, an Autolab 302N galvanostat/potentiostat with Pt counter electrode and Ag/AgCl reference electrode, and an illumination source of AM 1.5 G and 100 mW cm^{-2} were used.

Chronoamperometry (CA) and linear sweep voltammetry (LSV) measurements were performed to measure the photocurrent, dark current, and photocorrosion stability of the photocathodes. The scan rate and potential were set to 50 mV s^{-1} and 0 V_{RHE}, respectively, for both CA and LSV measurements. For photocorrosion stability tests, the photocurrent was measured at 0 V_{RHE} under continuous illumination. Nyquist and M–S plots were derived from EIS measurements, with

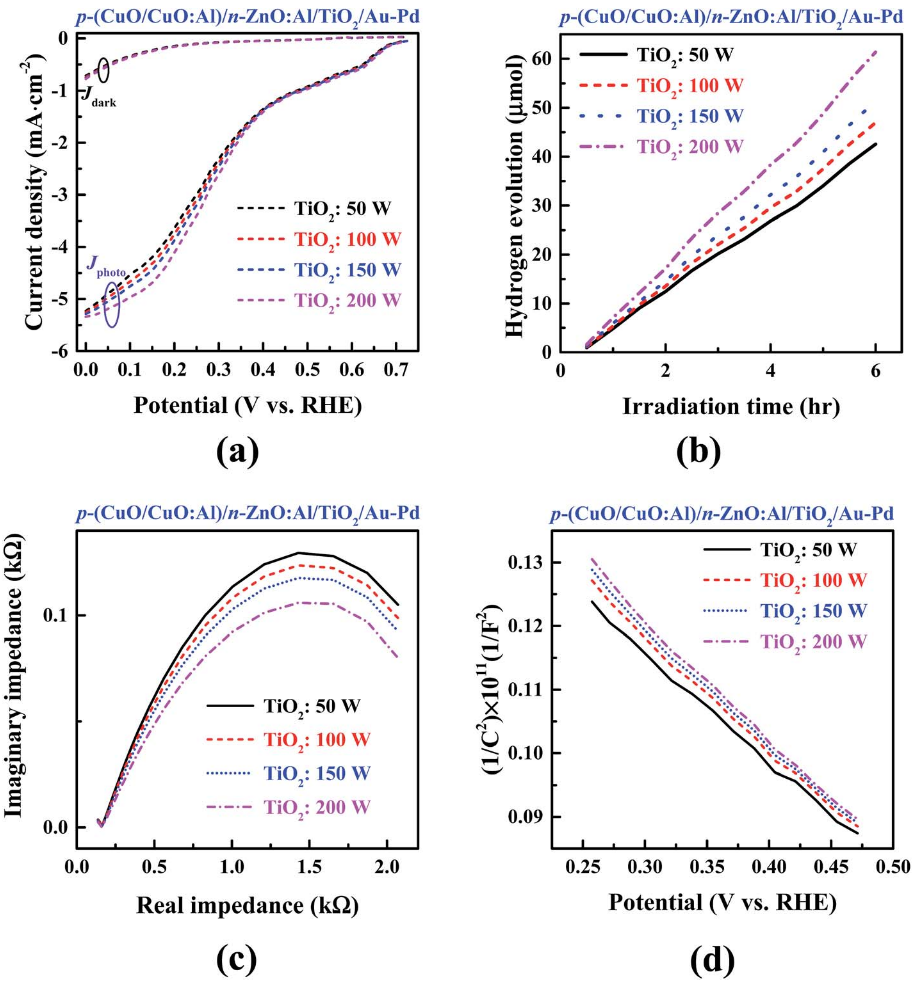

Fig. 14 (a) J–V, (b) hydrogen evolution, (c) Nyquist, and (d) M–S plots of the p-(CuO/CuO:Al)/n-ZnO:Al/TiO_2/Au–Pd photocathodes fabricated with TiO_2 sputtering power from 50 W to 200 W. Record-high J_{sc} of ~5.4 mA cm^{-2} at 0 V_{RHE} was achieved for the photocathode with TiO_2 sputtering power of 200 W.

Table 1 Default processing conditions

Layer	Thickness (nm)	Sputtering power (W): Metal oxide	Sputtering power (W): Al
CuO	30	25	—
CuO:Al	500	300	12
ZnO:Al	20	300	20
TiO_2	20	200	—

Table 2 Photocurrent and photovoltage of reported CuO photocathodes for PEC water splitting

No.	Photocurrent	Photovoltage	Ref.
1	−0.55 mA cm^{-2}@0 V_{RHE}	~0.35 V_{RHE}	25
2	−1.3 mA cm^{-2}@0 0 V_{RHE}	~0.35 V_{RHE}	26
3	−1.8 mA cm^{-2}@0 V_{RHE}	~0.42 V_{RHE}	27
4	−1.35 mA cm^{-2}@0 V_{RHE}	~0.45 V_{RHE}	28
5	−3.5 mA cm^{-2}@0 V_{RHE}	~0.4 V_{RHE}	39
6	−1.8 mA cm^{-2}@0 V_{RHE}	~0.35 V_{RHE}	40
7	−2.3 mA cm^{-2}@0 V_{RHE}	~0.3 V_{RHE}	41
8	−0.23 mA cm^{-2}@0 V_{RHE}	~0.15 V_{RHE}	61
9	−0.0045 mA cm^{-2}@0 V_{RHE}	~0.42 V_{RHE}	62
10	−1.5 mA cm^{-2}@0 V_{RHE}	~0.45 V_{RHE}	63
11	−1.1 mA cm^{-2}@0 V_{RHE}	~0.45 V_{RHE}	64
12	−2.5 mA cm^{-2}@0 V_{RHE}	~0.2 V_{RHE}	65
13	−2.3 mA cm^{-2}@0 V_{RHE}	~0.25 V_{RHE}	66
14	**−5.4 mA cm^{-2}@0 V_{RHE}**	**~0.53 V_{RHE}**	**This work**

Nyquist plots obtained by applying a 10 mV AC signal over a frequency range of 1 mHz to 100 kHz in the open-circuit potential condition, and M–S plots obtained at a fixed frequency of 5 kHz. Because the existence of dissolved O_2 would result in the reduction of oxygen rather than the production of H_2 as the dominant cathodic reaction, argon (Ar) gas was bubbled for 1 hour through the system before all PEC measurements.

The following equation was used to convert the measured potential with respect to the Ag/AgCl reference electrode ($V_{Ag/}$

$_{AgCl}$) to the potential on the reversible hydrogen electrode scale (V_{RHE}):[20]

$$V_{RHE} = V_{Ag/AgCl} + 0.197 + 0.059 \times pH \quad (3)$$

Optical transmittance, reflectance, and absorption of the fabricated samples were measured by a double-beam Shimadzu UV-3101 UV-vis-NIR scanning spectrophotometer. A high-resolution Philips CM300 TEM system was used to investigate the microstructural and interface properties of the fabricated samples, while a VG ESCALAB 220i-XL XPS system with a monochromatic AlKα source (1486.6 eV) was used to analyze their surface chemical composition. XPS data fitting for peak deconvolution was carried out using Gaussian and Lorentzian line shapes with Shirley background subtraction. XRD was performed using a Bruker AXS General Area Detector Diffraction System (GADDS) in θ–2θ scan with CuKα (λ = 0.15418 nm) radiation.

Conflicts of interest

We do not have any conflict of interest.

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